

Kim, Sang Hoon

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Careers

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| [4] Advanced Technology and Research Center, Vieworks | Jan. 2021 – |
| Research Topic: Image Sensor Beyond the Visible
Surface Enhanced Raman Scattering for Point of Care Testing | |
| Advisor: Dr. Yoon, Kyung Hun | |
| [3] Research Center, Samsung Display | Aug. 2018 – Jun. 2019 |
| Research Topic: Pixel Design for Under Display Camera | |
| Advisor: Dr. Kim, Gun Hee | |
| [2] M.S. in Electrical Engineering, KAIST | Sep. 2016 – Aug. 2018 |
| Research Topic: Photodoping of Graphene to Modulate Plasmonic Resonance Frequency | |
| Advisor: Prof. Jang, Min Seok | |
| [1] B.S. in Electrical Engineering, Ajou University | Mar. 2012 – Aug. 2016 |
| Concentration: Electromagnetics and Optoelectronics | |
| - Graduation with Top Rank and One Year Early | |
| - Military Service Leave of Absence | |
| | Dec. 2012 – Sep. 2014 |

Journals

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| [1] Efficient Photodoping of Graphene in Perovskite–Graphene Heterostructure [link] | 2019 |
| S Kim ⁺ , SG Menabde ⁺ , MS Jang [*] | Adv. Electronic Materials |

Conferences

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| [3] Quantum Dot-Based Direct X-Ray Detector for Mammography [link] | 2026 |
| S Kim, D Kim, K Heo, AG Shulga, V Goossens, K Yoon [*] | SPIE Medical Imaging |
| [2] Negative Refraction Dielectric Metasurface for High Spatial Resolution X-Ray Detector | 2025 |
| S Kim, K Heo, K Yoon [*] | Photonics Conference |
| [1] Efficient Photo-Doping of Graphene in Perovskite–Graphene Heterostructure | 2018 |
| S Kim ⁺ , SG Menabde ⁺ , MS Jang [*] | NANO Korea |

Patents

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| [5] Beam Steering Based High Spatial Resolution Indirect X-Ray Detector | 2025 |
| S Kim, K Yoon | Application in Progress |
| [4] PbS Quantum Dot Based Sensitive Direct X-Ray Detector | 2025 |
| S Kim, J Jeon | Application in Progress |
| [3] Method and apparatus for measuring surface-enhanced raman scattering [link] | 2024 |
| S Kim, K Yoon | WO2025170222A1 |
| [2] Method and device for detecting surface-enhanced raman scattering [link] | 2023 |
| S Kim, K Yoon | KR20240115636A |
| [1] Display panel and display device [link] | 2019 |
| S Kim, S Lee, G Kim, D Kim, S Moon, J Jeon, S Hong, T Yang | US12144223B2 |

Hands-on Skills

Interest Field of Study	Electromagnetics (Advanced), Device Physics (Advanced), Image Processing (Advanced)
Programming Language	Python (Advanced), MATLAB (Intermediate), C++ (Intermediate)
Simulation Tools	COMSOL Multiphysics (Intermediate), FDTD (Intermediate), MCNP (Intermediate)
Empirical Tools Set-Up (Instrument + Algorithm)	SWIR Microscope, Raman Spectroscopy, Laser Scanning with Galvano Mirror Photodetector & Image Sensor Evaluation System
Empirical Experience	Photodetector Modelling, Design, Fabrication, Evaluation, Analysis